Practical No. 05

ZOO 1203 - General Entomology

Body systems of an insect

Objectives:

After completing the practical you should be able to;

- Dissect an insect to identify the internal systems
- Identify parts of the digestive system, nervous and reproductive system of an insect
- Predict the feeding mode of an insect by looking at the mouthparts

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Dissection

Hold the specimen (Cockroach) with your left hand and clip the wings. Fix the specimen in a dorsal position on a dissecting tray with the help of pins passing through abdominal sterna and coxa of legs. Cut the lateral membrane (pleura) between the terga and sterna of the thorax and abdomen with a pair of fine scissors.

Posteriorly the two incisions should meet at the hindmost end of the abdomen. Proceed forward up to the anterior end of the thorax.

Give a transverse incision along the anterior border of the first thoracic segment and carefully remove the terga. The thoracic and the abdominal cavity are exposed. Put clear water in the tray. Remove fat bodies and tracheae to expose internal organs.

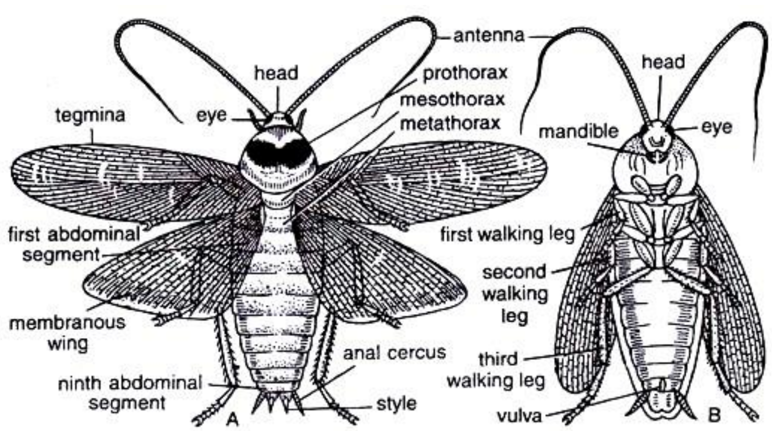
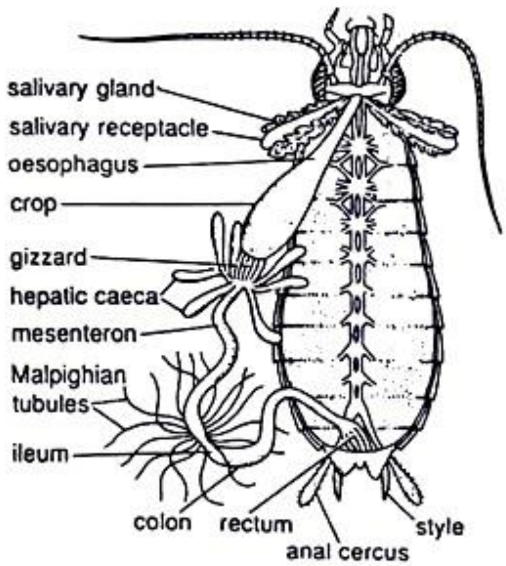


Fig. 6.1 : Cockroach (Periplaneta americana) A. Dorsal view, Male. B. Ventral view, Female

The Alimentary System

Carefully uncoil the intestine and stretch the alimentary canal to one side. Prevent it from coming back to the original position by pushing down a pin in the wax between the gut and the specimen.



: **Periplaneta sp.** Alimentary system

Mouth:
Ventral in position, located at the base of the buccal cavity.

Buccal cavity

An ill-defined chamber, bounded anteriorly by epipharynx and labrum; posteriorly by hypo-pharynx and labium; laterally by two mandibles.

Pharynx:
A short, vertically oriented tube opening into the oesophagus.

Oesophagus:
A short tube running posteriorly in the thorax.

Crop:
There is no demarcation between the oesophagus and the crop. In fact, the crop is a large sac-like dilatation of the oesophagus. It extends into the abdomen.

Gizzard or proventriculus:

A round, thick walled muscular structure, posterior to the crop. It has two parts, the anterior contains six chitinous teeth in the inner wall and the posterior two circular hairy cushions.

Mesenteron or mid gut:

A narrow tube running from the gizzard to the hind gut. 7 to 8 hepatic caeca are present at the junction of the gizzard and mid gut.

Proctodaeum or hind gut:

A narrow tube, divisible into 3 zones – ileum, colon and rectum. The junction of the mid and hind gut is marked by 60 to 70 extremely fine, yellowish Malpighian tubules (The tubules are excretory in function.). The rectum opens through the anus.

Salivary glands:

Two in number. The glands and the receptacles lie on the dorsolateral aspects of the crop. The ducts of the glands and receptacles run forward by the sides of the crop. The ducts from the two glands unite and those from the receptacles also unite to form two common ducts, which again unite and give rise to an efferent salivary duct opening on the ventral side of the hypo-pharynx.

The Nervous System

Fix the head of the specimen by pinning through the mandibles. The rest of the body should be fixed to the wax of the dissecting tray in the way already described. Carefully remove the epicranial plate of the head capsule and expose the cerebral ganglia.

Cut the pharynx and pull it out with the oesophagus. Remove the viscera and the ventral nerve cord is exposed. Expose the roots of the circumoesophageal connectives on the lateral sides of the brain and trace them to the points where they meet the sub-oesophageal ganglia.

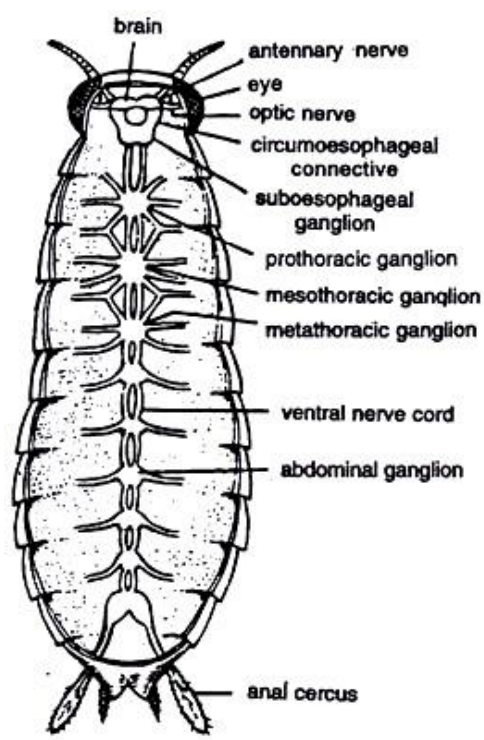
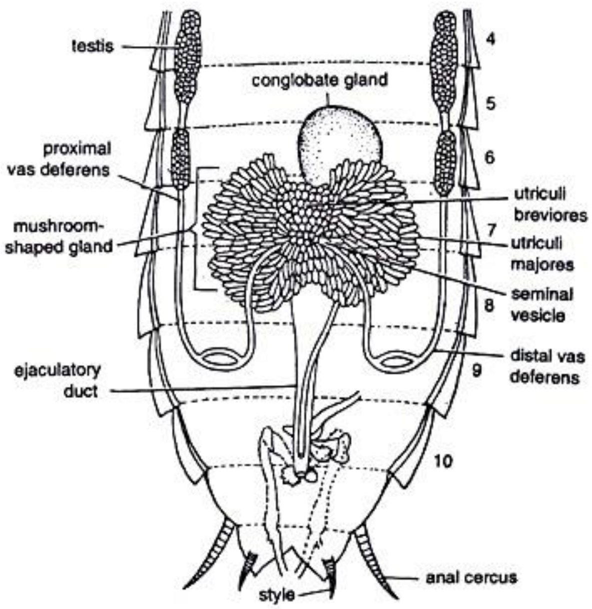


Fig. 6.4 : Periplaneta sp. Nervous system

The Reproductive System:

The sexes are separate. A pair of anal styles are present in male cockroach.

Male Reproductive System



: **Periplaneta sp.** Reproductive system. Male

Care should be taken while removing the terga. The paired testes are situated just beneath the 4th and 5th abdominal terga. The colour of the testes blends with the colour of the surrounding fat and it is difficult to distinguish them from the latter.

Testes:

Paired, small in size, made up of a number of vesicles.

Vasa deferentia:

Two narrow tubes. Arising from the testes run posteriorly and join to form the muscular ejaculatory duct.

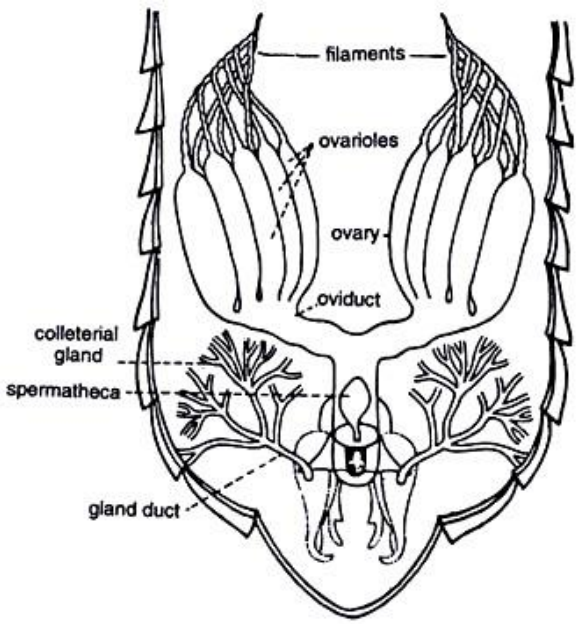
Ejaculatory duct:

It opens in a median genital armature, opening through male gonopore.

Conglobate gland:

Median, ventral to the ejaculatory duct and associated with it.

Female Reproductive System



Periplaneta sp. Reproductive system. Female

Ovaries:

Paired, located under the 4th and 5th abdominal terga. Each ovary has 8 tubules, the ovarioles, which are held together.

Oviducts:

Two, short, wide tubes. The ovarioles unite to form the oviduct. The two oviducts join posteriorly and form a muscular tube, the vagina.

Vulva:

It is a slit-like external opening of the vagina.

Colleterial glands:

A pair of much branched glands opening on either side of the vulva.

Spermatheca:

A median sac opening into the genital pouch.

Activities and questions

1. Dissect the given specimen and draw a complete diagram of the systems you can identify.